

#5 12-bits

2's complement binary representation

1)

-13

$$\begin{array}{r|l}
 2 & 13 \\
 \hline
 2 & 6 \rightarrow 1 \\
 2 & 3 \rightarrow 0 \\
 2 & 1 \rightarrow 1 \\
 & 0 \rightarrow 1
 \end{array}
 \begin{array}{r}
 000000001101 \\
 111111110010 \\
 + \\
 \boxed{111111110011}
 \end{array}$$

2)

56

$$\begin{array}{r|l}
 2 & 56 \\
 \hline
 2 & 28 \rightarrow 0 \\
 2 & 14 \rightarrow 0 \\
 2 & 7 \rightarrow 0 \\
 2 & 3 \rightarrow 1 \\
 2 & 1 \rightarrow 1 \\
 & 0 \rightarrow 1
 \end{array}
 \begin{array}{r}
 \boxed{00000011000} \\
 \text{Get 10 12 bit 2's complement.}
 \end{array}$$

3)

-1

Get to 2's complement then take inverse and add 1.

$$\begin{array}{r|l}
 2 & 1 \\
 \hline
 & 0 \rightarrow 1
 \end{array}
 \begin{array}{r}
 000000000001 \\
 111111111110 \\
 + \\
 \boxed{111111111111}
 \end{array}$$

5)

2048

$$100000000000$$

There is overflow. The largest positive number that can be represented in 12-bit 2's complement is 2047.

The binary number shown for 2048 starts with 1 indicating it is negative, but if you add a 0 it will be more than 12 bits.

4)

-2048

$$\begin{array}{r|l}
 2 & 2048 \\
 \hline
 2 & 1024 \rightarrow 0 \\
 2 & 512 \rightarrow 0 \\
 2 & 256 \rightarrow 0 \\
 2 & 128 \rightarrow 0 \\
 2 & 64 \rightarrow 0 \\
 2 & 32 \rightarrow 0 \\
 2 & 16 \rightarrow 0 \\
 2 & 8 \rightarrow 0 \\
 2 & 4 \rightarrow 0 \\
 2 & 2 \rightarrow 0 \\
 2 & 1 \rightarrow 0 \\
 & 0 \rightarrow 1
 \end{array}
 \begin{array}{r}
 100000000000 \\
 011111111111 \\
 + \\
 \boxed{100000000000}
 \end{array}$$

by adding 0s

For the negative numbers, first get them to 2's complement and then take the inverse of each 1 or 0 and then add 1.

For the positive numbers, they are just put in 2's complement and 12 bit form.